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Teaching Young Children to Make Bids to Play to Peers With Autism Spectrum Disorder

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Abstract

Peer-mediated intervention (PMI) within the inclusive early childhood setting increases social exposure between children with autism spectrum disorder (ASD) and typically developing peers. However, PMI research to date has provided limited data on the adherence of strategies taught to typically developing peers during training. The present study examined the extent to which a packaged intervention consisting of video modeling (VM) was an effective tool to promote bids to play by typically developing peers to children with ASD during unstructured indoor and outdoor play. A multiple probe across participants design demonstrated that all three typically developing peers learned to invite children with ASD to play after observing video models and children with ASD increased independent responses to initiations with least-to-most prompting from an adult.

Keywords

early childhood inclusion, peer-mediated intervention, video modeling

Inclusive educational settings provide children with autism spectrum disorder (ASD) with access to typically developing peers and models of social behavior. Because impairments in social communication and social interaction are two core characteristics of ASD (American Psychiatric Association, 2013), access to social interaction with peers is often identified as a high priority for service providers and parents of children with ASD (Pituch et al., 2011). However, access alone often fails to produce social interaction (Gresham, 1984). Typically developing peers generally require some form of training to interact with children with ASD, as they may need to respond differently than they would with their typically developing peers (e.g., initiate more frequently or be more persistent; Ferraioli & Harris, 2011; Laushey & Heflin, 2000; Odom, 2000).

Peer-Mediated Intervention (PMI)

PMI is an evidence-based practice for improving social outcomes among children with ASD that teaches typically developing peers to initiate interactions with children with ASD (Paynter et al., 2017; Whalon et al., 2015; Zagona & Mastergeorge, 2016). Extensive single-case experimental studies show gains in social interaction among children with ASD and their typically developing peers when PMI is used (Chang & Locke, 2016; Gunning et al., 2019; Wang et al., 2011; Watkins et al., 2015; Zhang & Wheeler, 2011). In addition, recent randomized controlled studies show improved social status and decreased isolation among individuals with ASD as a result of peer training (Kasari et al., 2012).

PMI can yield positive outcomes even when introduced at very young ages. Thiemann-Bourque and colleagues (2016) taught seven typically developing preschool peers to prompt the responses of children with ASD using the Picture Exchange Communication System (PECS®). Peers were taught during four, 30-min sessions to *Stay-Play-Talk* (English et al., 1997) to children with ASD using role-play, feedback, prompts, and reinforcement. The researchers demonstrated that typically developing peers could mediate the delivery of prompts using a least-to-most prompting hierarchy. The outcomes are promising in that children at a very young age (i.e., preschool) can mediate successful social interactions with their peers with ASD, which could alter the trajectory of social behavior among children with ASD before they reach the primary school level.

Despite the efficacy of PMI, there are several aspects of the intervention that are unknown to researchers and may be difficult to replicate due to a lack of data reporting

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typically developing peers' implementation fidelity. In fact, most published studies of PMI indicate that adults deliver prompts and reinforcement to typically developing peers as part of the intervention procedures, yet fail to specify how much adult involvement is required (Gunning et al., 2019; Kern & Aldridge, 2006). This issue of unknown adult involvement may be particularly relevant when PMI is administered at the preschool level (e.g., Thiemann-Bourque et al., 2016), as younger children likely need extensive support when learning to interact with peers with ASD. The omission of data on independent implementation of PMI by typically developing peers limits conclusions regarding the overall efficacy of the intervention and strategies for improving typical peer performance if and when needed. Furthermore, practitioners are not able to determine essential levels of adult involvement when planning to implement the intervention in their classrooms or clinics.

Some researchers have identified this limitation of PMI studies and have started incorporating strategies to increase independence among typical peers during the intervention process. For example, Hu et al. (2018) taught 13 typically developing peers to provide visual and physical prompts, modeling, and reinforcement to children with ASD during a play intervention to improve social interaction among children with ASD. During training, the researchers used role-play activities and ensured that typically developing peers met predetermined mastery criteria before the intervention was initiated. During intervention, researchers continued to collect data on whether typically developing peers successfully and independently prompted children with ASD as part of a larger procedural integrity (PI) checklist. Collecting PI of typical peers during PMI is important as they are the primary intervention agent. Research could expand upon these initial efforts by explicitly evaluating peer fidelity as a dependent measure and, if necessary, generating procedures to improve implementation fidelity of typical peers during PMI.

Video Modeling (VM) for Typically Developing Peer Training

VM is a cost-effective and evidence-based intervention strategy that reduces adult-delivered prompts, allows for reliable modeling of target behaviors, and supports generalization (Bellini & Akullian, 2007). Although variations exist, VM is used to teach a variety of new skills to individuals with and without disabilities. VM typically involves the following: A child views a target behavior on a video screen, an interventionist replicates the conditions of the video (e.g., materials, scenarios), and an interventionist asks the child to imitate what is seen in the video (Bellini & Akullian, 2007). Although VM is an evidence-based method for teaching social behaviors to children with ASD (Qi et al., 2018), it has been minimally evaluated as a strategy to teach typically developing peers to initiate interactions with children with ASD. However, it has been used effectively to teach children with ASD to initiate play interactions with peers (Nikopoulos & Keenan, 2004). Given its cost-effectiveness, practical features (e.g., reliable modeling of target behaviors), and reduced need for ongoing prompting, VM may be a useful tool for teaching typically developing peers to initiate play with children with ASD.

A small body of research suggests VM may promote extended play interactions between children with ASD and their typical peers without the need for ongoing prompting from adults (Dueñas et al., 2019; MacDonald et al., 2009; Maione & Mirenda, 2006). In these studies, the same video model was presented to *both* a child with ASD and a typically developing peer that modeled play actions and play statements during those interactions. Across the aforementioned studies, results indicate positive outcomes for children with ASD, with varied outcomes on independent social interaction by typical peers. Additional research is needed to determine whether VM is an effective tool for teaching typically developing peers a set of behaviors necessary for initiating and responding to play interactions of children with ASD.

Despite the efficacy of PMI for increasing social interactions between children with ASD and their typically developing peers, very few studies have examined the extent to which typically developing peers carry out skills acquired during PMI (e.g., prompting, initiations, continuations) following initial training or if adult-delivered prompts play a primary role in ensuring typically developing peers mediate social interactions (Gunning et al., 2019).

Additional data are needed to evaluate whether current peer training practices ultimately lead to effective and reliable implementation of PMI (Zagona & Mastergeorge, 2016). Capturing the PI of typical peers' implementation during PMI is an important area for research as accuracy of intervention implementation may reduce the need for adult facilitation. VM has potential for teaching typically developing peers to interact with individuals with ASD (Dueñas et al., 2019; MacDonald et al., 2009), though additional research is needed to determine the reliability of VM for doing so. Therefore, the purpose of the current study was to address the following research questions:

- **Research Question 1:** What are the effects of VM on typically developing peer's adherence to the steps to initiate a play interaction with a child with ASD?
- **Research Question 2:** What are the effects of the intervention package (e.g., peer training, least-to-most prompting, and reinforcement) on children with ASD's acceptance of play invitations?
- **Research Question 3:** Do the effects of the intervention package maintain beyond the intervention?

Method

Participants

Participants were recruited from an early childhood program affiliated with a Midwestern university. Three children diagnosed with ASD and three typically developing peers participated in the study. Children with ASD and typically developing peers were placed in dyads based on the lead general education teacher and director's discretion, as they were familiar with the characteristics of all children in the classroom.

Typically developing peers. The typically developing peers were enrolled in an early childhood education classroom for 4-year-old children where the children with ASD were also included. The director and general education teacher selected typically developing participants using the following criteria based on previous PMI research (Gunning et al., 2019; Zhang & Wheeler, 2011): (a) well-liked by peers, (b) positive social partner with the focal child, (c) generally compliant with adult directives, (d) able to attend to a task or activity for at least 10 min, (e) willing to participate, and (f) able to attend school on a regular basis. The children's parents provided consent to participate in the intervention and were in support of their child's role in the inclusive classroom setting. The three typical peer participants were 4 years old and demonstrated age-appropriate social and developmental skills according to their lead teacher based on the Child Observation Record (COR) Advantage, which is a valid observation-based assessment tool (Wakabayashi et al., 2019). Amber was a White female, Emi was an Asian female, and Milo was a White male. All three typically developing peers had been enrolled in the program for less than 1 school year at the start of the study.

Children with ASD. The participants with ASD were enrolled in an Early Intensive Behavioral Intervention (EIBI) program and were eligible based on an outside diagnosis of ASD from an approved autism evaluation center within the state. Participants with ASD spent 3 hr of their school day in a general education early childhood classroom and 4 hr per day in an intensive applied behavior analysis classroom. Participants with ASD were selected for inclusion in the present study because the researchers observed that the children did not respond to peer initiations and did not initiate interactions with peers. Regina was an Asian female and was 4 years old at the start of the study. Enzo and Xane were both White males and were 4 years old at the start of the study.

Observations by the first author took place at five time points across 2 weeks to collect direct observation data in the general education classroom and provide information regarding children with ASD's social behavior in the inclusive classroom. In addition, participants were assessed on two standardized measures. The *Mullen Scales of Early Learning* (MSEL; Mullen, 1995) is a standardized assessment of development from infancy to 68 months of age. Developmental quotients (DQs) are calculated by dividing the subscale age-equivalent score by the child's chronological age and multiplying by 100 (Lord et al., 2006). The *Preschool Language Scales–5th Edition* (PLS-5; Zimmerman et al., 2012) is an observational measure used to assess the receptive and expressive language abilities of preschool age children. Prior to intervention, Regina's Mullen DQ score was 56.3, and her PLS-5 total language standard score was 103 (1st percentile). She imitated two-component actions and vocalizations inconsistently during play with adults and peers. Enzo's Mullen DQ score was 64.9 and PLS-5 total language standard score was 129 (1st percentile). He imitated two-component actions and vocalizations from adults only. He was able to parallel play with a peer with adult prompts. Enzo preferred to roam the room and engaged in high rates of vocal and motor stereotypy. Xane's Mullen DQ score was 56.4, and PLS-5 total language standard score was 109 (1st percentile). He imitated two-component actions and vocalizations both when requested and spontaneously. He was able to follow two-component vocal directions from an adult without prompts and engaged in parallel play with peers, yet demonstrated a preference for adult interaction.

Early childhood classroom staff. Three early childhood staff members participated in the study by presenting video models to typical peer participants. The lead teacher was a White female, 52 years of age, held a master's degree in education, and had 18 years of teaching experience. The assistant teacher was a White female, 22 years of age, held a bachelor's degree in education, and was in her first year of teaching. The pre-service student teacher was also a White female, 21 years of age, and was currently completing her last semester of studies before graduating with a bachelor's degree in education.

Behavior technicians. Three behavior technicians employed by the EIBI program provided least-to-most prompting to the children with ASD during the intervention. The behavior technicians were White females, age 21 to 23, and held a bachelor's degree in special education or psychology. This was their first year employed as behavior technicians.

Setting

Baseline, intervention, and maintenance sessions took place during child-directed play activities in two areas of the school, including the inclusive classroom during child-choice time and the outdoor space connected to the school during recess. The inclusive classroom enrolled 15 children and was taught by a lead teacher, assistant teacher, and a pre-service student–teacher majoring in child development

Table 1. The Task-Analyzed Steps to Make a Bid to Play and to Accept a Bid to Play.

Making a bid to play by typical peers	Children with ASD accepting a bid to play
1. Walk over to peer with ASD	
2. Taps peer with ASD on the shoulder	
3. Says name of peer with ASD using audible voice	1. Looks in the direction of peer when name is called
4. Waits for peer with ASD to look in their direction	
5. Extends hand to peer with ASD or hands a play item	2. Takes peer's hand or play item
6. Says, "Come play!" to peer with ASD	3. Follows peer to play/activity
7. Begins activity with peer with ASD	4. Engages in activity for 2 min

Note. ASD = autism spectrum disorder.

and education. The general education early childhood classroom included pretend play, large group, art, snack, sensory, blocks, library, and cognitive work. All areas of the room were open and available for children to explore with adult supervision during the hour-long child-choice time. Child-choice time in the classroom was 1 hr in length and recess was 30 min. The outdoor recess area included climbing equipment, green space, bikes, a large teeter totter, merry-go-round, sand box, and materials, including balls, bubbles, buckets, and shovels. Behavior technicians trained to deliver applied behavior analytic programming supported the children with ASD in the inclusive classroom.

Materials

Researchers used a handheld digital camera (*Hero 4 Go Pro*™) to film the video models and collect data, and *iMovie* to edit the content. Videos were loaded onto participant assigned iPads in an application called *My Pictures Talk* for viewing. Headphones were used to cancel potentially distracting noises in the classroom. The Board Certified Behavior Analyst® who supervised the behavior technicians and the children with ASD's programming along with the early childhood lead teacher, completed an informal preference assessment survey and activities that were enjoyable for both children in the dyad and were included in the *My Pictures Talk* application. Materials used for indoor play activities were commonly available in the inclusive classroom (e.g., bubbles, blocks, dress-up clothes). Recess materials were commonly available outside or taken outside daily in the classroom bag, which included two pool noodles, hula hoop, buckets and shovels for sand box, and bubbles. Children had access to any materials that were available at child-choice time in the classroom and during recess across all conditions of the study. Additional materials for staff training and during intervention included procedural steps sheet, pencil, and data sheets.

Video models. Based on reported activities of interest, 10 to 20 s videos were created using familiar male and female typical peers not involved in the study as models who played both the initiators and responders. Each video model depicted the task-analyzed bid to play and 3 to 5 s of the chosen indoor and outdoor activity. The indoor activity video models (i.e., eight in total) included musical chairs, pat-a-cake, puppets, dress-up, blocks, books, bubbles, and a game involving the London Bridge nursery rhyme. The outdoor recess video models included bubbles, London Bridge nursery rhyme game, chase, hula hoop, sand box, and pat-a-cake. Only activities preferable to both children in the dyad were included in the iPads. For instance, the iPad used with one dyad did not include a video for playing chase as one of the children in the pair was reported to not like that activity.

Dependent Variable and Dependent Measurement Systems

All sessions were video recorded using a handheld digital camera by either the first or second author. The mobility of the handheld digital camera allowed for ease of recording indoors and outdoors. Videos were used to record the primary dependent variable, and live data were used to record the secondary dependent variable. Behavior technicians collected dependent variables of interest using a paper data sheet in the classrooms.

Typically developing peer implementation fidelity. The primary dependent variable was typically developing peer implementation fidelity, which was defined as the percentage of steps completed correctly by typically developing peers to invite a child with ASD to play, as shown in the task analysis in Table 1. Typically developing peer implementation fidelity was summarized as the percentage of steps performed correctly out of a total of three performance sessions (i.e., seven steps per opportunity = 21 total steps). One of the sessions occurred inside the classroom during choice time and two during recess. The total steps completed correctly across three sessions were divided by the total possible steps, 21, to calculate the percentage of steps completed correctly by the typically developing peer.

Children with ASD acceptance of bids to play. The secondary dependent variable was the total number of steps completed correctly without adult prompting by the child with ASD (see Table 1). The total steps completed correctly across three sessions were divided by the total possible steps (12 steps in total, four steps per session) to calculate the percentage of steps completed correctly by the child with ASD.

Interobserver Agreement (IOA)

A sample of 33% of videos were randomly selected across participants and conditions (baseline, intervention, and maintenance) for IOA. The coding was conducted by the first and second authors. At the time of the study, both were doctoral students in special education and child development, and both held master's degrees in special education. In addition, the first author was a BCBA and the second author was enrolled in a BCBA certification program. The first and second authors scored dependent variables from videos and compared their data with the data collected live by behavior technicians to calculate IOA. The first and second authors scored the videos for occurrence of dependent variables. A step on the implementation checklist was scored as an agreement if there was an agreement on whether the step occurred and whether the response performed by each child was unprompted or prompted. Disagreement was scored when observers disagreed on the type of response that was performed. Agreement was calculated for each child in the dyad across conditions by dividing the number of agreements by the number of agreements plus disagreements and multiplying by 100 to yield a percentage. Agreement for typically developing peers was 97% (range = 90%–100%) for Amber, 91% (range = 75%–100%) for Emi, and 95% (range = 90%–100%) for Milo. Agreement for children with ASD was 87% (range = 75%–100%) for Regina, 93% (range = 85%–100%) for Enzo, and 91% (range = 83%–100%) for Xane.

Experimental Design

A multiple probe design (Ledford & Gast, 2018) across dyads was used to examine the effects of the VM intervention on typically developing peer's bids to play. Multiple probe design was preferred over multiple-baseline design to minimize potential issues that could occur due to failed bids to play or interactions between child participants without providing intervention. The study consisted of three phases: (a) baseline, (b) intervention, (c) and maintenance. Following the logic of a multiple-baseline design, intervention was staggered for each dyad. Baseline data were collected following the initial typical peer training phase, that is, training outside the classroom and the 2-week practice in the inclusive classroom. This was done to measure the effects of PMI alone before introducing VM for additional modeling and coaching. After baseline stability was demonstrated for all three dyads, treatment was introduced to the first dyad. When the first dyad demonstrated an increase in the level of performance from baseline, baseline probes were obtained for the remaining dyads and the treatment was introduced to the second dyad. This process was repeated for the third dyad. The multiple probe design allows for a demonstration of stable baseline with intermittent measurement of the target behavior. The multiple probe design was ideal, as typically developing peers were not previously observed to initiate bids to play with children with ASD, and it was expected that such a pattern would continue without intervention.

Procedures

Initial typically developing peer training. Prior to baseline, the typically developing peers were trained by the first and second authors using an adapted procedure outlined by Katz and Girolametto (2013). The children were brought into a room adjacent to their classroom for training. The first and second authors led the one-time 30-min training for typically developing peers. The training session involved the following: (a) teaching the peers the names of children with ASD in their classroom using photographs, and two of each child's favorite activities; (b) reading a children's book titled *Pete the Cat and the New Guy* (Dean & Dean, 2014) and a short discussion of themes in accepting differences in others and finding common ground; (c) showing a 15-s video clip that depicted two typically developing peers inviting another child to play using the task-analyzed sequence; and (d) providing opportunities for rehearsal and feedback until children demonstrated the seven steps (see Table 1) in sequence one time without adult prompting with a peer partner. On average, children practiced 4 times before achieving accuracy, which was 100% independent performance of the steps on the task analysis. The procedure was further adapted by conducting ongoing opportunities for typically developing peers to practice with children with ASD in the inclusive classroom rather than in a separate room. Staff presented one opportunity daily for 2 weeks. These sessions involved a staff approaching a typically developing peer during child-choice time and asking them to invite their friend to play (child with ASD). Typically developing peers were then provided with least-to-most prompting and specific feedback to complete the seven steps in the task analysis (see Table 1). If children refused, during these opportunities, staff were instructed to return in 5 min to make the request. If children refused again, staff were instructed to note the opportunity as a refusal.

Staff training. Immediately following baseline, the first and second authors trained the three early childhood staff and three behavior technicians on the VM procedure. A 2-hr

training was conducted using a behavior skills training model, where instruction, modeling, rehearsal, and feedback were provided (see Parsons et al., 2012). Following live modeling of the procedure, behavior technicians and early childhood staff were placed in groups to practice the procedure with the materials. Each staff member was given the opportunity to role-play with another adult, ask questions, and receive feedback from the PI checklist until mastery criterion of 90% accuracy was achieved. The staff were also given a self-monitoring checklist to reference as needed during the training and between intervention sessions. In addition, once implementation was initiated in the classroom, the first and second authors collected PI data, and provided corrective feedback during the first two sessions of implementation.

Staff PI. PI was obtained for teaching assistants who delivered the VM intervention to the typically developing peers on 33% of sessions. Data were collected by the first and second authors from video for the following steps during intervention: (a) approaches student with iPad and says, "What do you want to play with (child with ASD's name)?"; (b) allows child to select video model from iPad; (c) ensures child is watching video by redirecting them to look at video if they look away for more than 2 s; (d) says, "Go find (child with ASD's name) and invite them to play"; (e) does not provide any additional vocal instruction; and (f) provides specific feedback, "You asked your friend to play" or represents video model if Steps 1 to 3 in the task analysis are incorrect, and repeats Steps c to f. PI for staff working with Regina and Amber during baseline was 100%, during intervention was 86.5% (range = 72%–100%), and 100% for mastery probes. For Enzo and Emi, PI was 100% during baseline, 87% (range = 72%–100%) during intervention, and 100% during mastery probes. For Xane and Milo, PI was 100% during baseline, 95% (range = 81%–100%) during intervention, and 100% during mastery probes.

Baseline. After children had completed the initial training phase, that is, training outside the classroom and the 2-week practice in the inclusive classroom, typically developing peers were asked to use the skill they learned in training to invite their friends with ASD to play. The researcher approached the typically developing peer and reminded them to play with the child with ASD by stating, "Go find (child with ASD's name) and invite them to play." This was done once during indoor child-choice time and twice during recess. Interventionists did not deliver prompts or reinforcement during baseline. If typically developing peers refused, the interventionist stated they would return to ask again in 5 min. Once 5 min had passed, the researcher restated "Go find (child with ASD's name) and invite them to play." After two consecutive refusals, the session was terminated.

Intervention. Intervention sessions involved the presentation of the video model to a typically developing peer, once during indoor child-choice time in the classroom, and twice during recess outside. Sessions were conducted 3 days per week. Mastery criterion was set at 90% or higher across two sessions. A session began when an assistant teacher presented the menu of video activities on the iPad to a typical peer participant and asked, "What do you want to play with (child with ASD's name)?" The typically developing peer then selected an activity on the iPad by selecting the video corresponding to the activity. This part of the intervention was planned to motivate typically developing peers and decrease potential refusals.

After determining the activity, the teacher showed the video model twice, then instructed the typically developing peer to make the bid by saying, "Go find (child with ASD's name) and do what you saw in the video." No additional vocal instructions were provided at this time. If the child imitated the video model and the invitation and interaction occurred, the teacher gave specific positive feedback by saying, "You got (child with ASD) to play musical chairs with you." If the child did not imitate the first three steps modeled in the video or imitated incorrectly, the video was presented a third time with the directive, "Do what you saw in the video," and no additional vocal instructions. This was done to ensure the typically developing peer gained the attention of the child with ASD, as Steps 1 through 3 involved locating the child, tapping them on the shoulder, and stating their name. These were important steps to emphasize as children would often say, "Come play with me" without gaining attention of the peer with ASD during baseline. If the child continued to fail to respond to the video, the teacher was instructed to move on, but this did not occur. Once children were engaged in the activity, early childhood staff and behavior technicians remained in proximity and timed the 2-min play sessions. If children attempted to leave the area, they were physically or verbally prompted to remain in the activity.

Behavior technicians prompted the behavior of children with ASD using the system of least prompts (SLPs; Doyle et al., 1988) to respond to their peer and engage in the activity. Prompt topographies were specific to each child with ASD for each step in the task analysis but generally followed a prompt hierarchy from least to most intrusive: (a) gestural prompt, (b) vocal prompt, and (c) physical prompt. For each step in the task analysis, behavior technicians waited 5 s before delivering the least intrusive prompt; if the child with ASD did not respond, then the technician moved to the next most intrusive prompt. Behavior technicians delivered tokens to the child with ASD contingent on correct responding after each session; tokens were later exchanged for known reinforcers. All children with ASD regularly used token economy systems in the inclusive classroom, and tokens had been conditioned as reinforcers prior to participation in this investigation.

Maintenance. A week after intervention mastery criterion was met, a maintenance probe was conducted for each participant consisting of three opportunities (i.e., two during recess and one during choice time) over the course of 1 day. Each probe occurred 1 week apart. The maintenance probe was conducted similar to baseline, where typically developing peer was asked to invite their friend with ASD to play, except the typically developing peer was no longer exposed to the video model. The typical peer selected an activity based on what was available in the room at the time or activities that they recalled from video models. No additional prompts were administered to the typically developing peer or the child with ASD.

Results

Figure 1 shows the percentage of independent steps completed for all dyads during baseline, intervention, and maintenance. During baseline, Amber demonstrated an absence of independent steps for making bids to play with Regina. Upon introduction of the VM procedure, Amber demonstrated an immediate increase in the number of independent steps completed from zero during baseline to 28% on the first intervention session with data showing an increasing trend with some variability (range = 28%–100%). Regina's independent steps to accept the bid to play increased from zero during baseline to 17% on the first intervention session with data showing an increasing trend with some variability (range = 17%–100%). The dyad showed a rapid increase in level and trend in responding, with no overlap between baseline and intervention phases. Amber and Regina met mastery criteria on the 10th session. Maintenance probes demonstrated that Amber had maintained 81% of the steps of the bid for play and Regina had maintained 75% of steps, post intervention and in the absence of video models.

During baseline, Emi demonstrated an absence of independent steps completed for making a bid to play. Upon introduction of the VM procedure, Emi demonstrated an immediate increase in the number of steps completed from zero during baseline to 38% on the first intervention session (range = 38%–100%), and Enzo increased responding from zero during baseline to 25% on the first intervention session (range = 25%–100%). The dyad showed a rapid increase in level and trend in responding, with no overlap between baseline and intervention phases. Emi met mastery criteria on the fifth session and Enzo on the sixth session. Maintenance probes suggest Emi and Enzo had maintained 100% of the steps to make a bid to play and accept a bid to play, respectively.

During baseline, Milo demonstrated an absence of independent steps for making a bid to play. Upon introduction of the VM procedure, Milo demonstrated an immediate increase in percentage of steps completed from zero during baseline to 62% on the first session (range = 62%–100%). Xane also completed zero steps in baseline and an immediate increase to 60% on the first intervention session (range = 60%–100%). The dyad showed a rapid increase in level and trend in responding, with no overlap between baseline and intervention phases. The typically developing peer and child with ASD acquired the skill on the second session and met mastery criteria on the third session. Maintenance probes suggest Milo had maintained 95% of steps for making a bid to play and Xane had maintained 83% of the steps to accept a bid to play.

Tau-U Effect Sizes

A nonoverlap effect size metric, Tau-*U*, was calculated for dependent variables of interest (i.e., typical peers' bids to play and child with ASDs' acceptance of bids to play). Tau-*U* is a nonparametric effect size that yields the nonoverlap of data between baseline and intervention phases, while controlling for possible trends during baseline. The effect size can be interpreted as small (0.65 or lower), medium-to-large (0.66–0.92), and large (0.93–1.0) (Parker & Vannest, 2009; Rakap, 2015). A web-based Tau-*U* calculator (see http://www.singlecaseresearch.org/calculators/tau-u) was used. The weighted Tau-*U* for typically developing peers was 1.0, 95% confidence intervals (CIs) = [0.65, 1], $p < .001$, and the weighted Tau-*U* for children with ASD was 1.0, 95% CIs = [0.58, 1], $p < .001$. These metrics are consistent with visual analysis, as baseline data remained at zero across all participants.

Discussion

The present investigation suggests VM can be an effective approach for improving performance of typically developing young children to initiate bids to play to children with ASD in an inclusive classroom. In addition, children with ASD independently accepted bids to play and continued interactions with typically developing peers after least-to-most adult prompting. The study contributes to the PMI literature by assessing responding of typically developing peers before, during, and after two forms of peer-mediation training (see Watkins et al., 2015; Zagona & Mastergeorge, 2016). In addition, the study successfully trained early childhood staff and behavior technicians to implement the intervention using behavioral skills training. The findings show that researchers should consider assessing implementation of intervention strategies by typically developing peers following PMI training, as children may not be performing steps accurately following traditional peer training strategies (i.e., instruct, model, role-play, and feedback).

Research, Clinical, and Educational Implications

The study also extends the measurement procedures used by Hu and colleagues (2018) by making performance of

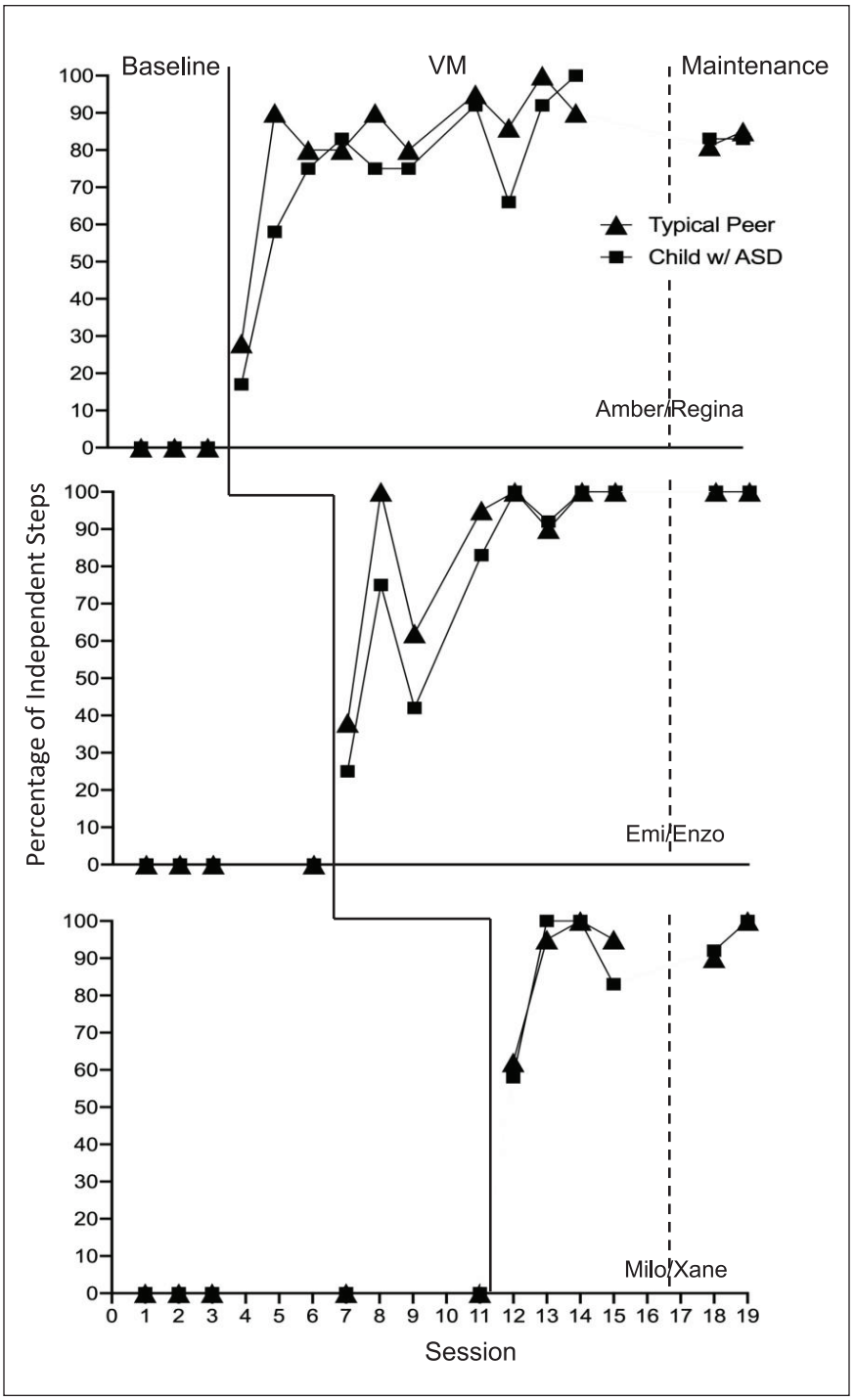


Figure 1. The percentage of independent steps to make a bid to play performed by typically developing peers in triangles and steps completed independently to accept a bid to play by children with ASD in squares.
Note. ASD = autism spectrum disorder; VM = video modeling.

typically developing peer implementation fidelity, the primary dependent measure. Teaching accurate performance to typically developing peers can be done prior to intervention by ensuring typically developing peers achieve mastery, as in Hu and colleagues (2018), but may also be required during training and after training to promote longer effects of the intervention. For example, an important outcome from the present investigation was that the

typically developing peers failed to demonstrate any of the specific procedural steps during baseline. This was surprising as all typically developing peers received an explicit training prior to baseline for involvement in PMI and involved typical peers achieving 100% accuracy in implementation and 2 weeks of ongoing coaching in their classrooms. This outcome has important implications for both research and clinical or educational practices. That is, the precise level of training required to achieve sustained change in behavior (e.g., bids to play) in typically developing preschoolers can vary significantly.

From a clinical or educational perspective, it is important to consider the lack of independent performance of typically developing peers during baseline, despite previous instruction in PMI. Conversely, during the VM intervention, all three typically developing peers demonstrated immediate improvement on performance and quickly met the mastery criterion, that is, in four to 10 sessions. Amber transitioned to intervention first and reached mastery by the 10th session. Emi and Milo transitioned sequentially following Amber and reached mastery in eight and four sessions, respectively. VM was an effective and efficient intervention for improving the independent performance of typically developing peers to engage in bids to play with peers with ASD.

From a research perspective, monitoring typically developing peer implementation fidelity during PMI is essential for the internal validity of a study. To replicate the study, researchers must be able to clearly identify all of the components within an intervention and the amount of those components implemented with accuracy. For example, it is insufficient to report that adults provided prompts to typically developing peers during PMI without specifying the type and number of prompts needed in order for typically developing peers to perform sufficiently in their roles as interventionists. By documenting the extent to which peers implement essential features of PMI following certain types of training (i.e., preliminary PMI training, followed by VM), the present investigation strengthens PMI literature by carefully describing the independent variable at the level of the typically developing peer and the extent of adult involvement.

All participants continued to perform the bids to play with above 80% accuracy 2 weeks after VM was terminated. The persistence of typically developing peers in engaging in interactions with children with ASD is important, as it creates ongoing social interaction and learning opportunities for children with ASD (Goldstein et al., 2007). However, in the present investigation, the video model served as the signal to initiate an interaction. As such, when the video was removed, the teachers instructed children to approach their peers with ASD (i.e., stating, "Go ask your friend to play"). Although no prompts were provided following that instruction, we cannot say that children were completely independent in approaching and interacting with peers during the follow-up measurement probes due to the initial verbal instruction provided by their teacher. An important objective for future research is to assess whether typically developing peers engage in spontaneous initiations with peers with ASD following PMI and VM training.

This study also used combined preferences (i.e., preferred activities for children with ASD and typically developing peers) to enhance benefit for typically developing peers to serve as playmates to children with ASD. PMIs are often implemented during unstructured, child-led activities where typically developing children may need to be interrupted from a preferred activity to invite another child to play or accept an invitation to play (Goldstein et al., 2007). Change in performance of typically developing peers in this study may have been a result of reinforcement in the form of access to preferred activities or direct feedback from their teacher. In the case of combined preferences, it is important to note that the activities were readily available to the students and access was not denied outside of the study. When designing PMI, it is important to consider natural reinforcement contingencies like toys and activities preferred by both typically developing children and children with ASD and social attention to increase the likelihood that desired social behaviors will persist over time.

In the present study, VM was delivered across two settings (i.e., recess and indoor child-choice time) and typically developing peers learned to initiate bids to play with peers with ASD in both environments. Programming generalization is a recommended strategy for promoting novel responding (Stokes & Baer, 1977). Future research could assess whether bids to play generalize to additional settings not programmed during instruction.

Limitations

Although the findings of this study promote the use of VM for training typically developing peers, a few important limitations should be considered. First, typically developing peers acquired the steps to make bids to play very quickly (i.e., in one to two sessions on average). It is clear that the VM intervention changed the behavior of typically developing peers, yet it consisted of a number of components (i.e., child-choice and adult-delivered social praise following interaction). Future research might involve introducing these components gradually and assessing change in behavior of typically developing peers, such as in an add-in component analysis (Ward-Horner & Sturmey, 2010). In addition, in this case, the VM intervention was used to achieve reliable performance of a skill already in repertoire. Future research should examine whether VM is as effective with children who have not been introduced to and demonstrated an ability to perform the target skills.

Furthermore, the current investigation did not assess how interactions were maintained post bids to play. Although VM was an effective strategy for ensuring accurate initiations of bids to play by typically developing peers, adult prompting was needed to sustain engagement in open-ended activities for children with ASD. This study did not examine the extent to which children with ASD responded to further initiations of peers or initiated to peers to sustain interactions following the initial bid. Play was observed as relatively brief interactions that included closed-ended (e.g., musical chairs, social motor games like pat-a-cake) and open-ended (e.g., bubbles, books, art, puppets) play activities. Given the effectiveness of VM to increase play initiations and for teaching play skills to children with ASD (Bellini & Akullian, 2007), it may have been beneficial to show a video model of children engaged in an extended play routine to both children at the same time (see Dueñas et al., 2019; MacDonald et al., 2005, 2009). The use of data collection systems to measure changes in the duration of interactions as well as engagement level of children could evaluate qualitative dimensions of treatment effectiveness in PMI.

Although follow-up data were collected after the last intervention session for each dyad, it is unknown whether the intervention impacted long-term changes in the social environment. Qualitative data such as interviews of children and teachers along with observations extending past 2 weeks could allow future researchers to evaluate whether increased interactions impacted social relationships in a meaningful way (Goldstein et al., 2007). Future PMI studies would benefit from collecting social validity data. Although this study did not formally incorporate social validity measures, anecdotal reports suggest that the intervention was not disruptive to classroom activities and was valued by teachers. Indirect and observational measures of social validity from classroom teachers, peers, and participants may provide information regarding the observed changes in social relationships or opportunities as well as evaluate the acceptability of the intervention. Although, during maintenance probes, typically developing peers were able to perform the bid to play without VM exposure, they were still prompted or asked to invite their peer with ASD to play. Therefore, the initiations of typically developing peers were not fully independent. An important future direction of this research is not only to look at accuracy of performance but ensure the behavior is occurring without adult involvement, which is what we hope will occur in inclusive early childhood classrooms during unstructured time. Furthermore, a major limitation of this study is that we did not collect generalization data. Although we programmed for generalization by teaching in more than one environment, we do not know whether bids to play transferred to other environments or other peers.

Finally, this intervention was a package and components of the package were not evaluated for their individual effects on participant behavior. Therefore, we are unable to state which components were essential for effecting change. In addition, we did not collect PI data for one of the intervention components, least-to-most prompting and reinforcement procedures delivered by the behavior technicians to children with ASD. This is an important limitation as acceptance of bids to play by children with ASD may have heavily relied on these procedures and we cannot speak to the accuracy of intervention delivery without these data.

Conclusion

As previously noted, this study provides an intervention procedure that incorporates child-choice by allowing typically developing peers to select toys and activities from a menu rather than being directed to an activity or toy. The combined list of preferences provided an easy to use, informal preference assessment that practitioners can use to evaluate features of the intervention that may influence typically developing peers' performance. In this investigation, we explicitly measured peer behavior, which allowed us to examine whether peers accurately initiated interactions with children with ASD. This study addresses limitations in previous peer-mediated research in terms of ensuring implementation fidelity for typically developing peers. The results also show that child participants with ASD learned to accept bids to play in the inclusive early childhood setting and the initiations of typically developing peers maintained following the removal of VM across settings (i.e., indoor child-choice and recess) and generalized to children with ASD not in the original training dyad.

Authors' Note

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